Weekly Progress Report – Lightweight Linux GUI Distros for Small-Scale Private Cloud Deployment

This report summarises the progress completed during Week 1 and Week 2 of *Examination of Lightweight Functional Linux GUI Distros for Small-Scale Private Cloud Deployment project*. The purpose of this report is to demonstrate weekly continuity, practical implementation, and baseline testing progress.

Week 1 – Research and Environment Setup

During Week 1, research was conducted on private cloud computing, virtualization technologies, and lightweight Linux operating systems. VMware was selected as the virtualization platform for all practical testing activities. Three lightweight Linux distributions were selected for comparison: Lubuntu, Xubuntu, and Linux Lite. These distributions were chosen due to their reputation for low system resource usage while still providing graphical desktop environments.

Virtual machines were configured using identical specifications to maintain fair testing conditions across all systems. Each VM used Ubuntu 64-bit as the operating system type, 2 GB RAM, 2 processor cores, 20 GB virtual storage, and NAT network configuration.

The operating systems were installed using the “Erase disk and install” option in order to simplify partitioning and maintain consistency throughout the setup process. Lubuntu and Linux Lite installed successfully without major issues. During the initial installation of Xubuntu, a different partitioning option was selected, which resulted in the operating system failing to boot correctly after installation. The virtual machine was recreated and Xubuntu was reinstalled using the “Erase disk and install” option, after which the system functioned normally.

During the Linux Lite installation, a message appeared requesting the removal of the installation medium before rebooting. After disconnecting the ISO image and restarting the VM, the operating system booted successfully.

Initial observations suggested that Xubuntu required a longer installation process compared to the other distributions. Later baseline testing showed that Lubuntu achieved the fastest startup time, while Linux Lite demonstrated second highest memory usage but significantly slower startup behaviour.

Week 2 – System Configuration and Baseline Testing

During Week 2, all three Linux distributions were updated and configured identically in preparation for baseline performance testing. Monitoring tools including htop and fastfetch/neofetch were installed to collect hardware and system usage information. Baseline measurements focused on idle RAM usage, boot times, storage utilisation, and general responsiveness. Testing was performed after allowing each operating system to remain idle for several minutes following startup.

Lubuntu recorded approximately 958 MB of idle RAM usage with a startup time of 13.047 seconds. Xubuntu recorded approximately 998 MB of idle RAM usage with a startup time of 15.610 seconds. Linux Lite recorded the second highest memory usage at approximately 992 MB, although it also recorded the slowest startup time at 42.072 seconds.

Disk usage information and additional system details were collected using df -h and fastfetch/neofetch. Screenshots of htop, fastfetch/neofetch, and desktop environments were captured as practical evidence for inclusion within the final IEEE paper.

The testing completed during Week 2 established measurable baseline performance data that will be used later in the project when deploying private cloud services and comparing system efficiency.

Current Project Status

At the end of Week 2, all selected Linux distributions have been successfully installed, configured, and benchmarked under identical virtual machine conditions. Baseline measurements and installation observations have been documented, providing a strong foundation for the next stage of the project.